Probe-Prüfung 2

1. a) $\varphi(\underbrace{4860}_{2^2\cdot 5\cdot 3^5}) \overset{(I)}{=} \underbrace{\varphi(2^2)}_{\overset{\|(III)}{\underbrace{(2^2-2)}_{2}}} \cdot \underbrace{\varphi(5)}_{\overset{\|(II)}{4}} \cdot \underbrace{\varphi(3^5)}_{\overset{\|(III)}{(\underset{243}{3^5}-\underset{81}{3^4})}} = 1296$

$(I)\ \varphi(m\cdot n) \overset{ggT(m,n)=1}{=} \varphi(m)\cdot\varphi(n)$
$(II)\ \varphi(p) = p-1$
$(III)\ \varphi(p^n) = p^n - p^{n-1}$

Euler: $ggT(a,n)=1$
$a^{\varphi(n)} \equiv 1 \mod n$

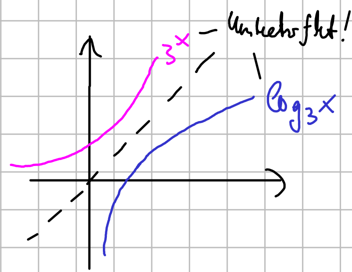
b) $2^{980} \mod 243.$

Euler: Voraussetzung $ggT(2, \underbrace{243}_{3^5}) = 1$ ✓

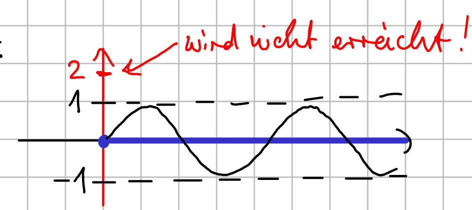
$\varphi(243) = \varphi(3^5) = 3^5 - 3^4 = 162 \overset{\text{Euler}}{\Longrightarrow} 2^{162} \equiv 1 \mod 243.$

Insgesamt: $\underline{2^{980}} = 2^{162\cdot 6+8} = (\underbrace{2^{162}}_{\equiv 1})^6 \cdot 2^8 \equiv 2^8 = 256 \equiv \underline{13} \mod 243$

c) $\log_3(x)$ hat $D =]0,\infty[= \mathbb{R}^+$



d) $\sin : [0,\infty[\longrightarrow \mathbb{R}$ nicht surjektiv:



e) $1 \cdot \sin(2x)$
Amplitude
Periode $\frac{2\pi}{2} = \pi$

2. $A = \{\text{"a"}, \text{"aa"}, \text{"b"}, \text{"inf"}\}$

a) $R = \{ (\text{"a"},\text{"a"}), (\text{"a"},\text{"aa"}), (\text{"a"},\text{"b"}), (\text{"a"},\text{"inf"}),$
$\cancel{(\text{"aa"},\text{"a"})}, (\text{"aa"},\text{"aa"}), (\text{"aa"},\text{"b"}), (\text{"aa"},\text{"inf"}),$
$(\text{"b"},\text{"b"}), (\text{"b"},\text{"inf"}),$
$(\text{"inf"},\text{"inf"}) \}$

b) reflexiv, da Diagonale in R, d.h. $\forall s \in A: (s,s) \in R$ ✓

nicht symmetrisch, da $(\text{"a"},\text{"aa"}) \in R$, aber $(\text{"aa"},\text{"a"}) \notin R$ ✗

antisymmetrisch: $(s_1, s_2) \in R \wedge (s_2, s_1) \in R$ ist nur für Diagonale erfüllt!

Hier ist $s_1 = s_2$ ✓

transitiv: $(s_1, \underbrace{s_2) \in R \wedge (s_2}_{\text{Überlappung}}, s_3) \in R$,

d.h. s_1 steht im Telefon vor (oder „=") s_2 $\wedge$ s_2 steht im Telefon vor (oder „=") s_3

$\Rightarrow$ s_1 steht im Telefon vor (oder „=") s_3, d.h. $(s_1, s_3) \in R$ ✓

[ODER: alle Überlappungen durchgehen!]

c) keine Äquivalenzrelation, da nicht symmetrisch ✗

Ordnung: reflexiv, antisymmetrisch & transitiv ✓

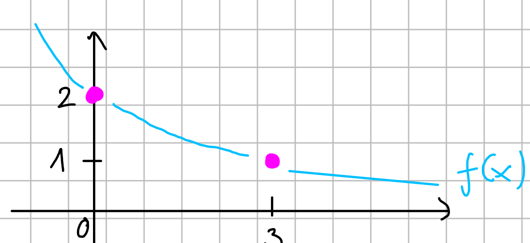
3. $f(x) = a\, e^{bx}$:

$(0;2)$: $2 = f(0) = a \cdot \underbrace{e^{b \cdot 0}}_{1} = a$

$(3;1)$: $1 = f(3) = a \cdot e^{b \cdot 3} = 2 \cdot e^{b \cdot 3} \overset{:2}{\Rightarrow} \frac{1}{2} = e^{3b} \overset{\ln(\dots)}{\Rightarrow} \ln\left(\frac{1}{2}\right) = 3b$

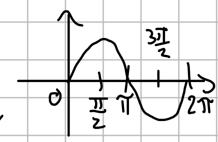
$\overset{:3}{\Rightarrow} b = \frac{\ln(1/2)}{3} = \frac{\ln(1) - \ln(2)}{3} = \frac{-\ln(2)}{3}$

also $\boxed{f(x) = 2 \cdot e^{-\frac{\ln 2}{3} x}}$



4. $f(x) = \underset{\uparrow \text{ Amplitude}}{2} \sin(3x - \frac{\pi}{6})$

3 ← Periode $\frac{2\pi}{3}$

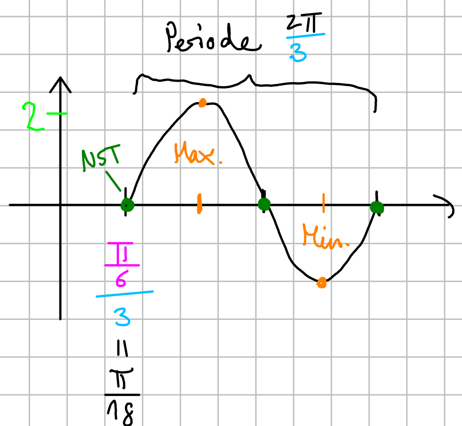


Max: Argument $3x - \frac{\pi}{6} = \overbrace{\frac{\pi}{2} + k \cdot 2\pi}^{\sin(x) \text{ wird maximal}} \Rightarrow x = \frac{2\pi}{9} + k \cdot \frac{2\pi}{3} \quad (k \in \mathbb{Z})$

Min: Argument $3x - \frac{\pi}{6} = \overbrace{\frac{3\pi}{2} + k \cdot 2\pi}^{\sin(x) \text{ minimal}} \Rightarrow x = \frac{5\pi}{9} + k \cdot \frac{2\pi}{3} \quad (k \in \mathbb{Z})$

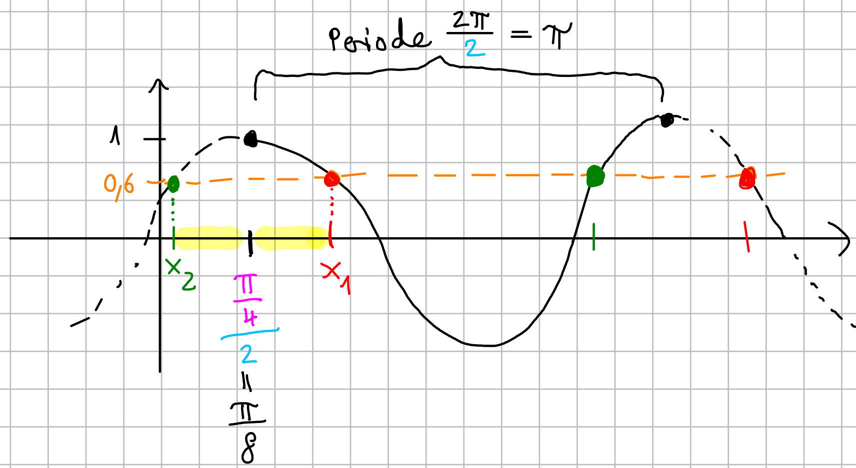
NST: Argument $3x - \frac{\pi}{6} = \overbrace{0 + k \cdot \pi}^{\text{NST von } \sin(x)} \Rightarrow x = \frac{\pi}{18} + k \cdot \frac{\pi}{3} \quad (k \in \mathbb{Z})$

ODER: graphische Lösung:



...

5. $1{,}6 = \cos\left(2x - \frac{\pi}{4}\right) + 1 \overset{-1}{\iff} \cos\left(2x - \frac{\pi}{4}\right) = 0{,}6$



x_1: $\quad 2x - \frac{\pi}{4} = \arccos(0{,}6) \implies x = \frac{\arccos(0{,}6) + \frac{\pi}{4}}{2} = x_1$

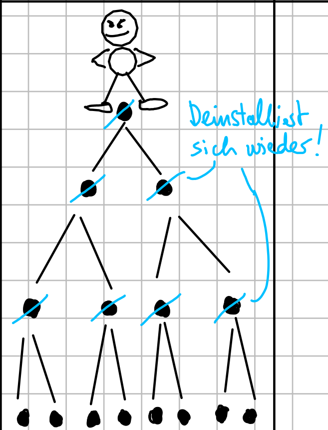
x_2: $\quad x_2 = \frac{\pi}{8} - \left(x_1 - \frac{\pi}{8}\right) = 2\cdot\frac{\pi}{8} - x_1 = \frac{\pi}{4} - x_1$

Alle Lösungen $\boxed{x = x_1 + k\cdot\underset{\text{Periode}}{\pi} \quad \text{oder} \quad x = x_2 + k\cdot\pi = \frac{\pi}{4} - x_1 + k\cdot\pi \quad (k\in\mathbb{Z})}$

[Probe mit x_2: $\cos\left(2x_2 - \frac{\pi}{4}\right) = \cos\left(\underbrace{2\cdot\left(\frac{\pi}{4} - \frac{\arccos(0{,}6) + \frac{\pi}{4}}{2}\right)}_{\frac{\pi}{2} - \arccos(0{,}6) + \frac{\pi}{4}} - \frac{\pi}{4}\right)$

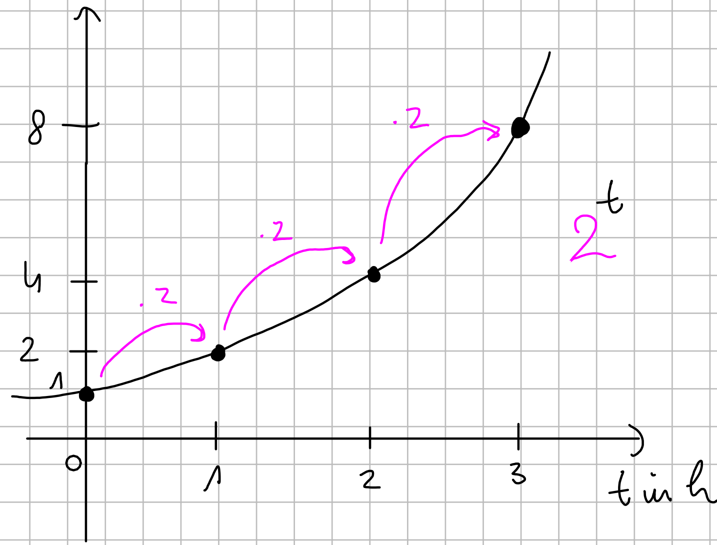
$= \cos\left(\frac{\pi}{2} - \arccos(0{,}6)\right) = 0{,}6$]

6.



t	$n(t)$
0	1
1	2
2	4
3	8
⋮	⋮

$\cdot 2$ $\cdot 2$ $\cdot 2$



a) $n(t) = 2^t$.

b) $1\,d = 24\,h$: $n(24) = 2^{24} = \underline{16{,}7\ldots \text{ Mio.}}$

c) Ansatz: $\underbrace{n(t)}_{2^t} \overset{!}{=} 7.200.000.000$ (Stand 2020: 7,79 Mrd.!)

$$\overset{\ln}{\Rightarrow} \ln(2^t) = \ln(7.200.000.000)$$

$$\Rightarrow t \cdot \ln(2) = \ln(7.200.000.000)$$

$$\Rightarrow t = \frac{\ln(7.200.000.000)}{\ln(2)} \approx \underline{32{,}75\,h}$$